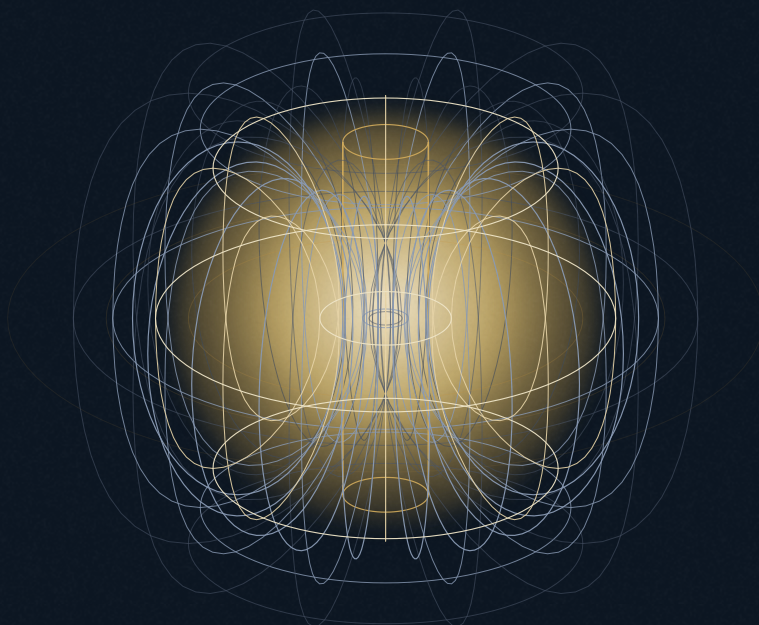




KRONOS FUSION ENERGY



PAPER 2.39 OF THE 2026 KRONOS SET

The Admissible Operating Window

Which Constraints Actually Bind

A breeder's operating point is boxed by competing constraints — and there is no free operating-point gain.

S_n ↔ centrepost-life invariant · q95/MHD floor, fluence ceiling · startup 1.7× margin

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

The Admissible Operating Window — Which Constraints Actually Bind

The operating window synthesized from two findings: a strict invariance — source strength and wall loading trade one-for-one against centrepost life across the whole density–temperature plane, so there is no free operating-point gain — and a coupled admissible-current window whose floor is the q95/MHD edge and ceiling the centrepost fluence, with the tritium-startup requirement non-binding at $\sim 1.7\times$ margin.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132231](https://doi.org/10.5281/zenodo.22132231) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords: [operating window](#) [source-strength invariance](#) [centrepost life](#) [constraint synthesis](#)

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
FOAK/NOAK/BOAK	first / n th / best of a kind
Hyperion	the ST breeder (internal: Mode D2)
Aegis	the generator, defense/installation housing (internal: Mode M)
MetroVolt	the generator, data-center housing (Mode M)

No free lunch in the operating plane

Where a breeder operates is decided by the constraints that box it in, not a single figure of merit. Across the whole operating plane, source strength and wall loading trade one-for-one against centrepost life — a fixed ratio, no corner where more neutron output comes free of centrepost cost.

THE BOX, AND WHAT BINDS

The admissible current is a coupled floor/ceiling problem: the floor is the q95/MHD edge, the ceiling the centrepost fluence, and the tritium-startup requirement — often assumed binding — sits inside with ~1.7× margin. So the true floor is the MHD edge, not fuel startup; the design current 9.66 MA sits inside.

Admissible operating window: a coupled floor/ceiling problem

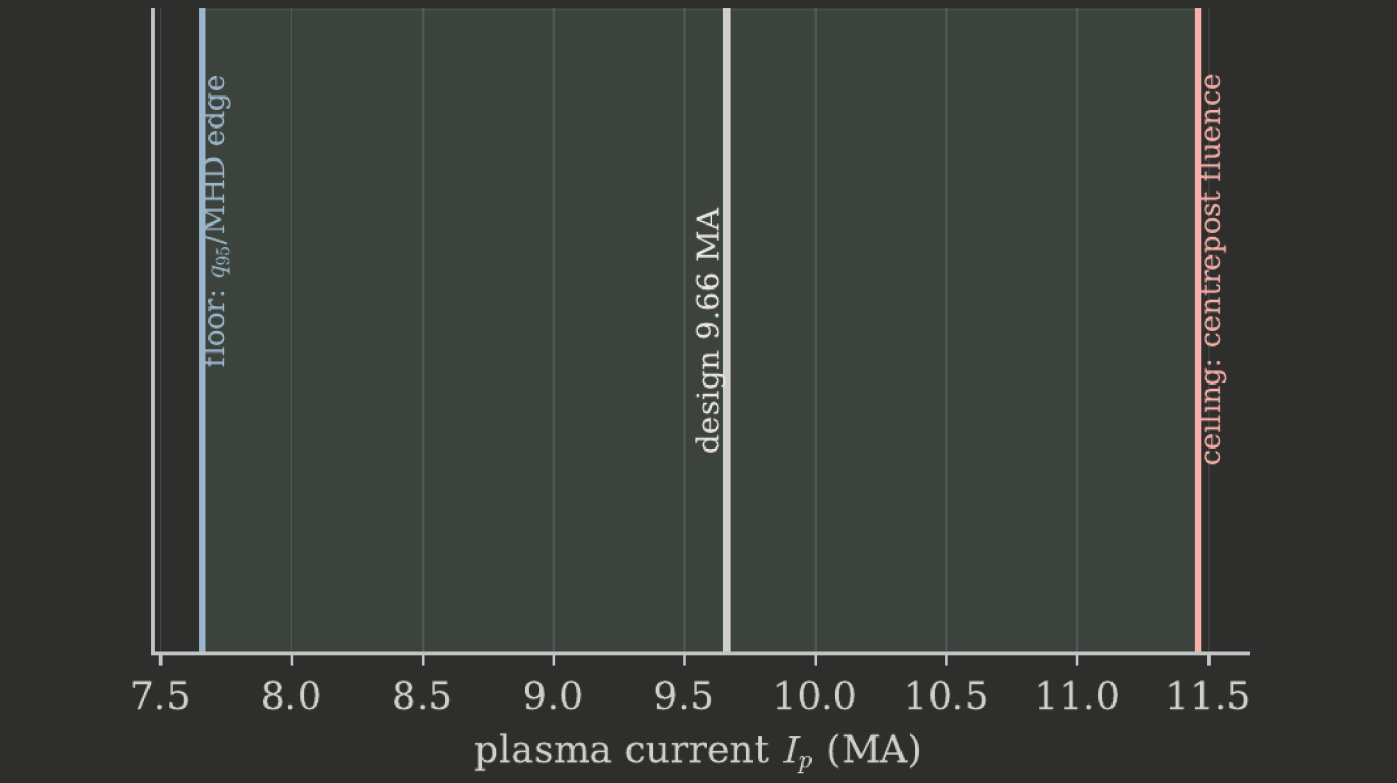


FIGURE 1. The admissible plasma-current window: q95/MHD floor, centrepost-fluence ceiling, design point inside (gate KX-L1-A21). CONSTRAINT MAP

§ 2

The actionable output

The synthesis identifies which constraints bind: the centrepost-fluence ceiling and the MHD-edge floor do; the tritium-startup floor does not. A designer moving the operating point trades centrepost life against edge stability, with fuel startup a non-issue in this window — a clearer picture than treating all constraints as co-equal.

SYSTEMS SYNTHESIS

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

FOUNDING PARTNERS — SCIENCE

Dr. Gerald Kulcinski

Co-Founder · Helium-3 & Advanced-Fuel Fusion
UW-Madison · 2023–Present

Dr. Carl Weggel

Chief Scientist / S.M.A.R.T. Design
MIT Alcator Program · 2022–Present

Dr. Robert J. Weggel

Magnetic Field Design
MIT Magnet Lab · 2022–Present

BOARD

Priyanca Ford

Founder & CEO · Executive Board
2022–Present

Bandel Carano

Finance / Strategy
Oak Investment Partners / Stanford · 2025–2026

Patrick Schweiger

Fusion Device Engineering
Oklo / CFS / TerraPower · 2025–Present

SCIENTIFIC ADVISORS

Dr. Steven O. Dean

Fusion Policy & History
DOE / Fusion Power Associates · 2025–Present

Dr. Patrick H. Diamond

Plasma Turbulence & Transport
UC San Diego · 2025–Present

Dr. Jack J. Dongarra

HPC & Numerical Algorithms
Turing Award · 2025–Present

Dr. Nasr M. Ghoniem

Chief Material Scientist
UCLA · 2025–Present

Dr. Siegfried Glenzer

Plasma Physics & Laser Fusion
Stanford / SLAC · 2022–Present

Dr. David A. Hammer

Pulsed-Power Fusion
Cornell · 2025–Present

Dr. Donald A. Spong

Plasma Theory & Confinement
ORNL · 2025–Present

Dr. Curtis Smith

Risk Assessment & Nuclear PRA
MIT / Idaho National Lab · 2025–Present

RADM (Ret.) David Goggins

Naval Engineering & Power-Plant Design
U.S. Navy · 2023–2026

Dr. Konstantin Batygin

Mathematician
Caltech · 2022–2025

Dr. Ruben Fair

Magnet Design / ITER Liaison

PPPL / Jefferson Lab · 2022–2025

Dr. Paul S. Weiss

Materials & Nanotechnology

UCLA · 2022–2025

SCIENTIFIC AUDITORS**Dr. Wilfred A. Cooper**

Plasma Equilibrium Theory

EPFL / Swiss Plasma Center · 2025–Present

Dr. Ahmed Hassanein

Plasma–Material Interactions

Purdue / Argonne · 2025–Present

Dr. Patrick E. Hopkins

Extreme Thermal Materials

U. of Virginia · 2025–Present

Dr. Peter Hosemann

Materials Joining & Structural Integrity

UC Berkeley · 2025–Present

Dr. Travis W. Knight

Advanced Nuclear Fuels & Systems

U. of South Carolina · 2025–Present

Dr. Philippe Lebrun

Cryogenics & Magnet Cooling

CERN · 2025–Present

Dr. Nitendra Singh

Nuclear Safety & Fuel Cycle

ITER · 2025–Present

Dr. Guido Van Oost

Magnetic Confinement & Fusion Systems

Ghent University · 2025–Present

Dr. Gary S. Was

Radiation Materials & Structural Integrity

U. of Michigan · 2025–Present

Dr. Ray Sedwick

Direct Energy Conversion

U. of Maryland · 2025

OPERATIONS**Michael Laughlin**

Chief Operating Officer

2022–Present

Martin Owens

Chief Strategy Officer

LANL / GE Hitachi · 2022–Present

MG (Ret.) Paul Pardew

Chief Contracting Officer

Army Contracting Command · 2022–Present

David Beck

Fusion Commercialization

U.S. Space Force · 2024–Present

Sushma Bhatia

Environmental & Legislative

Google · 2022–Present

Michael De Frenza

Technology Licensing

2023–Present

MG (Ret.) Robin L. Fontes

Cybersecurity & Defense

Army Cyber Command · 2022–Present

Vijay Gehani

Supply Chain & Components

INOX India · 2024–Present

Jon Michel Greenwood

IT & AI Infrastructure

Live Nation · 2023–Present

Brian C. O'Neill

National Security

CIA / ODNI · 2022–Present

Gen. (Ret.) Gustave F. Perna

DoD Liaison

Operation Warp Speed · 2022–2023

Andrea Romero

Marketing Lead

2022–Present

Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

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KRONOS FUSION ENERGY

The Admissible Operating Window

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kronosfusionenergy.com/physics-Validation-Simulation

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